

Before you build it, ask the people who'll use it.

An innovative solution that nobody needs is just an expensive hobby. This module is about the two techniques the study design names for finding out what users actually need — surveys and interviews — when to reach for each, and how to turn what people tell you into real requirements.

What this key knowledge covers

This summary distils everything you need for **KK06 — Before you build it, ask the people who'll use it.** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Breadth or depth — you can't have both from one method

Every data collection method trades **breadth** (how many people you hear from) against **depth** (how much you learn from each one). That single trade-off is the heart of this dot point, and it's why surveys and interviews exist as separate tools.

The survey net ☐

Think of a **fishing net** dragged across the whole school. It catches *a little* from *everyone* — fast, wide, countable. Priya surveys 600 students in a day and learns *how many* want vegetarian options. What she can't do is ask any one fish, "wait — why?"

The interview spotlight ☐

Think of a **spotlight** on one person. It reveals *a lot* about *a few* — slow, narrow, rich. Liam interviews the canteen manager for 30 minutes and uncovers the real bottleneck (EFTPOS refunds), something no checkbox would ever have surfaced.

A **survey reaches many people shallowly** and gives **quantitative, generalisable** data; an **interview reaches few people deeply** and gives **rich, qualitative** data. Name the trade-off — breadth vs depth — and you've earned the marks. Use the 3D scene above to see it: switch the instrument and watch how many people light up versus how much each one reveals.

Surveys — wide and countable

A **survey** is a fixed set of questions sent to many people, usually online (Google Forms, Microsoft Forms). Everyone answers the *same* questions, so the responses are easy to count,

chart and compare. Surveys are the go-to when you need to know **how many, how often, or which option is most popular** across a large group.

- **Breadth** — reach hundreds or thousands quickly and cheaply.
- **Quantitative** — closed questions produce numbers you can graph and compare.
- **Consistent** — everyone gets identical questions, so results are comparable.
- **Anonymous** — people answer honestly about sensitive topics.
- **Fast to analyse** — forms tools auto-tally and chart results.
- **Shallow** — you can't ask "why?" or follow an interesting answer.
- **No clarification** — a confusing question stays confusing; you get junk data.
- **Low response rate** — many ignore it, which can bias results.
- **Rigid** — you only learn about things you thought to ask.
- **Misreads** — people rush, click randomly, or misunderstand.

Priya posts a 6-question Google Form to all year levels. In two days, **612 students** respond. She learns: 71% would use a pre-order app, the worst queue is **12:55-1:05pm**, the most-wanted feature is "**see when my order is ready**", and 18% need allergen labelling. That's a clear, countable picture of *what most students want* — exactly what a survey is for.

Interviews — narrow and rich

An **interview** is a guided conversation with one person (or a small group) where the interviewer can **probe**, follow up, and read tone and body language. You learn *why*, not just *how many*. Interviews shine when you need depth: the workflow of an expert, the lived experience of a user with specific needs, or the messy reasons behind a behaviour.

Three flavours of interview

Fixed questions, same order, no improvising. Closest to a spoken survey — consistent and easy to compare, but rigid.

A planned set of questions *plus* freedom to follow up. The sweet spot for gathering needs — most school projects use this.

A free-flowing chat around a topic. Great for early discovery, hard to compare across people.

- **Depth** — probe with "why?", "tell me more", "what happens when...".
- **Qualitative richness** — stories, reasons, edge cases and emotions.
- **Adaptive** — chase unexpected answers a survey would miss.
- **Clarify on the spot** — fix misunderstandings immediately.
- **Empathy** — read tone and body language; build rapport.
- **Slow & costly** — minutes per person; you can only do a few.
- **Small sample** — can't generalise to the whole population.
- **Interviewer bias** — leading questions or tone sway answers.
- **Hard to analyse** — transcripts must be read and coded by hand.
- **Less anonymous** — people may hold back face-to-face.

Liam runs **four semi-structured interviews**: the canteen manager (Mrs Tran) and three students with food allergies. From Mrs Tran he learns the till can't easily refund a cancelled order — a hidden *non-functional* constraint. From a student with a severe nut allergy he learns that "vegetarian" labels aren't enough; she needs *per-ingredient* allergen info. No survey checkbox would have caught either. **Four people, but each answer reshaped the design.**

Side by side — and why you usually want both

Flip each card to reveal the contrast. The pattern is always the same: where a survey is strong, an interview is weak, and vice-versa. That's exactly why pairing them — survey for breadth, interview for depth — beats either alone. Researchers call this **triangulation**: two methods checking each other.

The method is only as good as the questions

You can pick the perfect method and still collect garbage if the questions are badly written. Two ideas to master: the **open vs closed** distinction, and the common **question faults** that quietly ruin your data.

Closed questions

Fixed choices — yes/no, multiple choice, a 1–5 rating. They produce **quantitative** data that's quick to count. Perfect for surveys.

"How often do you buy from the canteen each week? ☐ 0 ☐ 1–2 ☐ 3–4 ☐ 5+"

Open questions

Free text or free speech — the person answers in their own words. They produce **qualitative** data: rich, but slow to analyse. The lifeblood of interviews.

"What's the most frustrating part of buying lunch at school?"

Spot & fix the faulty questions

Each question below is broken in a classic way. Tap to reveal what's wrong and how Priya's team fixed it.

Who you ask decides what you learn

Collecting data isn't just choosing a method — it's choosing **who answers**. Two ideas decide whether your data is trustworthy: the **sample** (the slice of the population you actually reach) and the **response rate** (how many of them reply). Get these wrong and even a huge survey can mislead you.

Survey: confidence in 'how many' climbs as the sample grows (66 responses) — but every data point stays shallow. Great for breadth and numbers.

If GrazeGo only surveys the **queue at the canteen**, they miss every student who gave up and brings lunch from home — exactly the people the app could win back. A sample that doesn't represent the whole population produces **biased** data, no matter how many respond. Always ask: *who is missing from my sample?*

Which method fits the goal?

Drag each goal into the method you'd choose (or tap a chip then tap a bin on mobile). Some genuinely need **both**. When you're done, hit *Check*.

Tip on mobile: tap a chip to select it, then tap a bin.

Turning answers into requirements

Collecting data is step one. The *point* is to convert what users said into **solution requirements** — the bridge to KK05. A raw quote isn't a requirement; you have to translate it. Notice how each requirement traces back to a real piece of collected data.

From GrazeGo's data to its spec

Survey data tends to become **functional requirements and measurable non-functional targets** (numbers, popularity, thresholds). **Interview data** tends to surface **hidden constraints and edge-case requirements** that no checkbox would reveal. You need both kinds to write a complete specification.

Collecting data responsibly

The moment you collect data from real people, you take on responsibilities. Examiners love a question that asks you to weave an **ethical or legal** point into a data collection answer — so know these.

Tell people what the data is for, that taking part is voluntary, and that they can stop. For under-18s, you may need parent/guardian consent.

Don't collect names or identifying detail you don't need. This links to the **Privacy Act 1988** and the Australian Privacy Principles you met in Data Analysis.

Don't deceive participants or pressure them. Store responses securely and use them only for the stated purpose.

Neutral, non-leading questions and a representative sample aren't just "good practice" — biased collection produces unfair, misleading conclusions.

VCAA tie-in: the study design even asks "*how can the coordinator be sure every student submitted their survey — or that one student didn't submit 20?*" Data integrity (one response per person, verified) is part of collecting trustworthy data.

Decode what the question is really asking

Marks are lost when students *describe* for an *analyse* question. Tap each command word to see what it demands and roughly how many marks it signals.

Write answers that actually score

High-band answers follow a simple shape. **P-E-L**: make your Point, support it with Evidence from the scenario, then Link back to the user's need or the question. Try it below — the trainer watches for the moves examiners reward.

State which method and why, directly.

Cite the scenario — the users, the goal, the numbers.

Tie it to the need: breadth/depth, requirement, bias.

Practice prompt: "GrazeGo wants to know how many students would use a pre-order app. Recommend a data collection method and justify your choice." (3 marks)

VCAA-style questions · weak vs strong answers

Five questions, **19 marks** total. Read each, attempt it yourself, then open the weak answer (and *why* it loses marks), the strong answer, and the marking guide. Mark your own attempt —

the dial bottom-right tracks your total.

Distinguish between a survey and an interview as techniques for collecting user needs.

The GrazeGo team needs to know **how many** students across the school would use a pre-order app, and which features are most wanted. Recommend a data collection method and justify your choice.

Liam wants to understand the canteen manager's refund workflow and a nut-allergy student's experience reading food labels. Explain why an **interview** is more appropriate than a survey for these two goals, referring to the type of data each method produces.

A different team is building **RoamSafe**, an app to help international students find culturally familiar food near campus. They plan to survey only students queuing at the campus food court. Analyse **two problems** with this data collection plan and suggest how to improve it.

GrazeGo has three weeks and four team members. Propose a **complete data collection plan** that uses **both** a survey and interviews to determine user needs and requirements. Justify each method's role, and explain how the team would ensure the data they collect is **trustworthy**.

Everything in KK06, on one screen

Survey — same fixed questions to many people; *breadth, quantitative*, fast, cheap, anonymous; can't probe. **Interview** — guided conversation with a few; *depth, qualitative*, can probe, builds empathy; slow, small sample, hard to analyse.

"**How many / which / how often**" → survey. "**Why / how exactly / what's it like**" → interview. **Need the full picture / check a design** → both (triangulation).

Avoid **leading, double-barrelled, ambiguous, jargon/loaded** questions. **Closed** = quantitative (surveys); **open** = qualitative (interviews).

Representative **sample** (who's missing?), decent **response rate**, watch **bias** (self-selection, convenience), ensure **integrity** (one response each), gain **consent**, keep it **private**.

Surveys buy you breadth and numbers; interviews buy you depth and reasons — match the method to whether the question asks "how many" or "why", sample fairly, and turn what you hear into requirements.

Common traps & misconceptions

Exam trap

01 Listing strengths with no trade-off Writing "surveys reach lots of people" and stopping. Strengths only earn full marks when paired with a limitation or a comparison. ↳ Always answer breadth-vs-depth: more reach *but* less depth.

Exam trap

02 Ignoring the scenario Giving a generic textbook answer when the question describes a specific situation (an expert's workflow, a whole student body). ↳ Name the actual users and pick the method that fits *them*.

Exam trap

03 Confusing qualitative and quantitative Saying surveys give "detailed reasons" or interviews give "percentages of the whole school". You've swapped their data types. ↳ Survey = quantitative/breadth · Interview = qualitative/depth.

Exam trap

04 Forgetting sample & bias Recommending a survey but never mentioning who is sampled. A biased sample undermines the whole method. ↳ Add one line on a representative sample / response rate.

Exam trap

05 Treating raw data as the answer Stopping at "they collected the data" without converting it into requirements or a design decision. ↳ Show the link: this data → this requirement.

Exam trap

06 One method when the question rewards two An "analyse" or "recommend" question often wants you to justify combining methods. Picking only one caps your marks. ↳ Survey for breadth + interview for depth = triangulation.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Compare · 3 marks

Compare **interviews** and **surveys** as methods for collecting user needs.

✗ A weak answer

Interviews are talking and surveys are forms. Both collect data. (→ *Describes format only; no comparison of depth, scale or data type — 1.*)

✓ A strong answer

Interviews gather rich, detailed, mostly **qualitative** data and let the interviewer ask follow-up questions, but they are time-consuming and reach few people. **Surveys** reach many users quickly and produce **quantitative** data that is easy to compare, but the answers are shallower and you cannot probe unexpected responses. Interviews favour depth; surveys favour breadth.

How the marks are awarded

- 1 mark — interview = detailed/qualitative but small-scale.
- 1 mark — survey = large-scale/quantitative but shallower.
- 1 mark — an explicit comparison (depth vs breadth).

Q2. Justify · 2 marks

A team has two weeks to understand the needs of 400 potential users. Justify the most suitable collection method.

✗ A weak answer

They should interview everyone to get good data. (→ *Ignores the constraint — 400 interviews in two weeks is unrealistic — 0-1.*)

✓ A strong answer

A **survey** is most suitable. With 400 users and only two weeks, a survey can be distributed once and answered in parallel, producing comparable quantitative data at a scale that interviewing every user could never reach in the time available. A short round of interviews could follow to add depth if time allows.

How the marks are awarded

- 1 mark — selects survey.
- 1 mark — justifies it against the scale + time constraint.